

Com2uS Mobile Game Development

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During the middle of 2010, Com2uS wanted to create another mobile game targeted to the new era of the mobile phone game industry after the 2007 success of the Action Puzzle Family (APF) game for mobile phones. Mobile game users requested more attractive features in the new games, including being easy to learn, being easy to play, not requiring any previous knowledge, and being accessible anytime. In addition, the mobile games environment dramatically changed after the introduction of iPhone and Android devices. The new game Tiny Farm was designed to include the desired new features and also to cater to the new smartphone era. Given this new environment, Com2uS applied certain strategies from the waterfall and Agile software development process models that were appropriate for the development and release of this new game into the smartphone game market. Based on two mobile game development cases, students will learn the pros and cons of the waterfall and Agile development standard process models, in order to select a suitable development process for planning and designing modern mobile game software, and will be able to discuss the fast changing nature of mobile game industry.

In the late 2000s, the global mobile phone industry was shifting to a totally different paradigm following the successful launch of the iPhone by Apple. The iPhone allowed applications to be easily downloaded from the Internet so that even general users could become content providers and application developers for the mobile phone. Along with these developments, the expansion of WiFi and the long-term evolution currently in the 3G stage has reduced the cost of high-speed Internet close to zero. These broad environmental changes in the information technology (IT) industry further directed the technology trends of mobile games from standalone embedded games to mobile items downloadable from application stores such as the AppStore by Apple and Play Store by Google.

Freemium (Free-To-Play) is a pricing strategy by which games with simple features are provided free of charge, but money (premium) is charged for proprietary features, functionality or virtual goods (Hayes, 2008). Unlike premium games, users do not need to purchase free-to-play game titles in order to play them. In this sense, the freemium games are almost free. Instead of purchasing each game title, the freemium games generate the revenues during playing time by having users purchase items to enhance playing experience (i.e., In-App purchases). Free-to-play is a good example of mobile game monetization.

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The social network game (SNG) is a type of online game that is played through social networks, and it typically features multiplayer and asynchronous game play mechanics (Song, 2012). This category represents the most popular games played in the world, with many products hosting tens of millions of players. Because of the expansion of social network services, SNG is one of the popular game areas and represents the next evolution of earlier classic online games such as Starcraft, Warcraft and Dawn of War (Wikipedia, 2014c). As noted previously, SNG is a subset of online games but it dynamically interacts with social network services such as Facebook. FarmVille, Mafia Wars and The Sims Social (by Electronic Arts) are well known SNGs based on Facebook development platform (Zynga, 2014).

Many competitors exist for mobile games, and practically all game development companies are involved in the market (Kim, 2014). These include big gaming companies such as Electronic Arts and Disney as well as smaller, relatively new companies such as SundayToz and Zynga, which are all creating their own niches in the business. In the South Korean market, Next Floor was one of the new game development companies. Next Floor launched the mobile shooting game Dragon Flight, which is connected with Kakao (KakaoTalk is a multi-platform texting application that allows iPhone, Android, and BlackBerry users to send and receive messages for free). Next Floor grew to become one of the famous mobile game development companies in Korea. The game was a relatively simple shooting game like other shooting games, but—by attaching a service platform that adapts to the online social activities (competing with online social friends and also helping them)—has added significant value. Dragon Flight earned more than 250,000 U.S. dollars per day within less than 1 month (Kim, 2014).

Com2uS had been at the forefront of the mobile game industry since its inception in 1998. The firm built its reputation as the number one mobile games provider in Korea, which is one of the most advanced mobile game markets. Com2uS is a successful developer of many premium titles including Slice It!, Home Run Battle, and Tower Defense. It has successfully shifted to Freemium casual social games including Tiny Farm, Derby Days, and Magic Tree. The exceptional technology of Com2uS ranges from network expertise and content development to server operation and platform transition knowhow. The growing appetite for designing fun games on new technology has been the source of rapid and steady growth within the game industry, which led to its listing on the KOSDAQ in 2007 (Chung & Park, 2012; Com2uS, 2012; Kim, 2012). In September 2008, the mobile game division of Disney, mDisney, made a deal with Com2uS in which the latter would develop games for the former to be published and distributed in the United States. Disney Puzzle Family was the first game developed under the agreement, which was released later that month. The next title with Disney Interactive, Disney Game Parade, was released in the succeeding year. In 2008, Com2uS also began a game service on iOS platform by publishing Crazy Hotdogs and Chronicles of Inotia for iPhone and iPod Touch. In 2010, Com2uS began online golf game, Golf Star, for audiences in Europe and Korea. In April 2011, their game Slice It! recorded 10 million downloads on iOS and Android. In July 2012, the general manager from Com2uS iterated the vision, (*fun game development for all platforms*) of Com2uS during the Casual Connect in Asia Conference in Singapore (Lee, 2012). The conference has been produced by the Casual Games Association, and it is the premier event for the casual games industry with more than 5,500 professionals attending Casual Connect each year. Com2uS published some useful case studies for developing Freemium titles and shared their vision and experience of key success factors of Freemium games along with relevant statistics based analysis.

The idea of a casual game is that any user can play without any previous knowledge. Casual gamers are very commonly played in arcade game rooms and also with consumer electronic

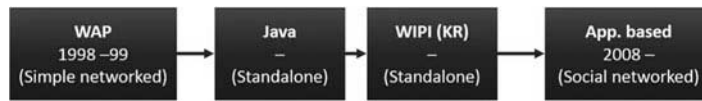


FIGURE 1 Mobile game trends in Korea.

devices such as personal computers, mobile phones, and game consoles. These types of smart-phone games are mostly premium games because they generate most of the revenues and users must purchase the games from application stores. *Slice It!*, *Homerun Battle*, and *Tower Defense* are flagship games from Com2uS. A premium game, however, can only generate revenue when it is installed because it is not based on value-added services but single product download purchasing.

In the late 1990s, wireless application protocol (WAP)-based mobile games were launched. Most games connected with mobile networks were WAP-based, but these did not click with very many users because the price of cellular data usage required to play these games were extremely expensive at the time. After Java programming platform for mobile (Java 2 Micro Edition [J2ME]) became available, more mobile games were developed based on Java, which required no network connections to play. As mentioned in the beginning, mobile games require a lot of data usage. Around 2008, many mobile games would also become aligned with social networks (Figure 1). Wireless internet platform for interoperability (WIPI) is a middleware platform driven by the South Korea government. This platform regulated that all mobile phones sold in Korea should implement the WIPI platform to avoid inordinate competition between mobile companies and to create a barrier to entry of foreign companies. But the regulation has been withdrawn since April 2009 (Wikipedia, 2014e).

Modern software development processes have evolved after the proliferation of the Internet (Iansiti & MacCormack, 1999). Although the companies of 1980s have different development processes, each process is designed to be flexible (Iansiti & MacCormack, 1999). Mobile game development processes have some similarities with the software development process during the early years of Internet era but required modifications to meet the needs of new mobile device users.

CASE RESEARCH METHODOLOGY

The case research project of Com2uS started on November 1, 2012 and completed on March 31, 2013. The project was targeted to produce two teaching cases that were suitable for technology management course. One case was designed for teaching the strategic decisions and technology trends of the mobile game industry (Kim, 2014). The other case was designed for teaching the software development management of the mobile games and this paper is the result of this case to understand the trends of mobile game software development.

This author visited Com2uS in December 2012 and interviewed relevant company managers and project leaders, including Mr. Youngil Lee, Chief Technology Officer, and Ms. B. Lee, a project leader. Concurrently, the questionnaire was sent via e-mail to Mr. S. Ryu (a lead producer of APF) and Mr. J. Lim (a lead producer of TinyFarm) seeking answers about the game

development processes used for APF and TinyFarm games. They were really supportive and even some of internal documents (marketing and development planning) were shared (but could not be available for public). All interviews (onsite and e-mail) were in Korean language and the quotes here on the following session are translated into English. Ms. Lee tried to facilitate better communication between the developers and the author. Unfortunately they did not have any detailed documentation of software developments (a characteristic and also a disadvantage of Agile development process) and also they were extremely busy in developing other game projects. So, most details of software development process were obtained from the responses to the questionnaires sent to Mr. Ryu and Mr. Lim (Appendix A).

CASES OF COM2US GAME DEVELOPMENT

This section describes two case studies of the actual mobile game development.

Action Puzzle Family (APF) Case

Puzzle Family brings twitch game action and excitement in the form of delightful classical puzzle games. All puzzles in the set are smooth and easy but loaded with excitement. Action Puzzle Family (APF) is known as one of the more popular Freemium casual games (Com2uS, 2014a). This game has been available on both Android and iOS platforms and has more than six million users. Users can try to get all the silly costumes for 20 comical characters and may compete with their friends to get higher rankings in eight delightful puzzle games. The goal of the game is simple: collect all puzzle pieces to move the family into a house with wacky features and each one of the eight eccentric puzzle games has a different theme (Table 1).

Development of APF started in 2006 with one producer (Mr. Ryu) and two software developers (Mr. G. Kim and Mr. J. Choi) that lasted till the end of the project. The budgeted development time was 10 months. In the beginning, there were a total of six members in the development team that included three software developers, one quality engineer, and one binary build (build code executable on Android platform from Java code) engineer. It was observed during the interview that some other members including—at least one graphic designer and one additional software

TABLE 1
Puzzle Games in Action Puzzle Family (APF)

<i>Game Name</i>	<i>Descriptions (Theme)</i>
Flick and Fly	Someone messed up the room. Clean it up before mom comes home.
Bong Bong	Blocks spring up to stick and pop! Have fun bouncing them around.
Copy Copy Cat	Remember the puzzle to duplicate it!
Who's Next	Waiting in line . . . How far can you count?
Order the Order	Too many orders to take! To keep them in order, you need speed.
Shanghai Pop	Match and pop the Chinese way.
Tap on Time	Tick tock, goes the clock! Tap on time to clear the blocks!
Hide and Seek	Baby puzzles hide: roll them around, and match them correctly.

developer—were involved in this game development but were not shown on the development planning documents. The producer, Mr. Ryu, and a lead software developer, Mr. Choi, were fully involved in the project from beginning to end. Other project members were temporary—they were in and out when the project needed certain types of features or functions that capitalized on their expertise. Since the development process of APF was still controlled by a small group (less than 10 people in total), it was more flexible than the general Com2uS development process (Figure 2). Some elements of the development process were not strictly adhered to. All elements of the process are enumerated here:

- *Development start*: Ideas of new games, based on classic puzzle games, were proposed.
- *Prototyping*: The task force team created a prototype of the game that could easily demonstrate its basic concept. In the case of APF, it took less than a month to make the first prototype game.
- *Alpha test (demo)*: The APF for the trial prototype was opened to limited channels such as mobile operators in Korea (SK Telecom, KTF, and LGT). Internally, the quality assurance team was actively involved to ensure stability and quality of the game. It took 3 months to ensure that APF was of top quality. It took 2.5 months from the development to starting alpha testing (the prototyping is in the middle between two phases).
- *Beta test (demo)*: Beta testing was primarily meant for fixing bugs, and customer service team of Com2US was actively involved during this phase. Balancing the game levels and difficulty were also finalized based on the alpha test done in the previous stage. The game would have been terminated in this phase if it was not good enough for the current market.
- *Commercial launch*: After completing beta testing, APF was launched on phones operated by SK Telecom (SKT) on August 11, 2007.

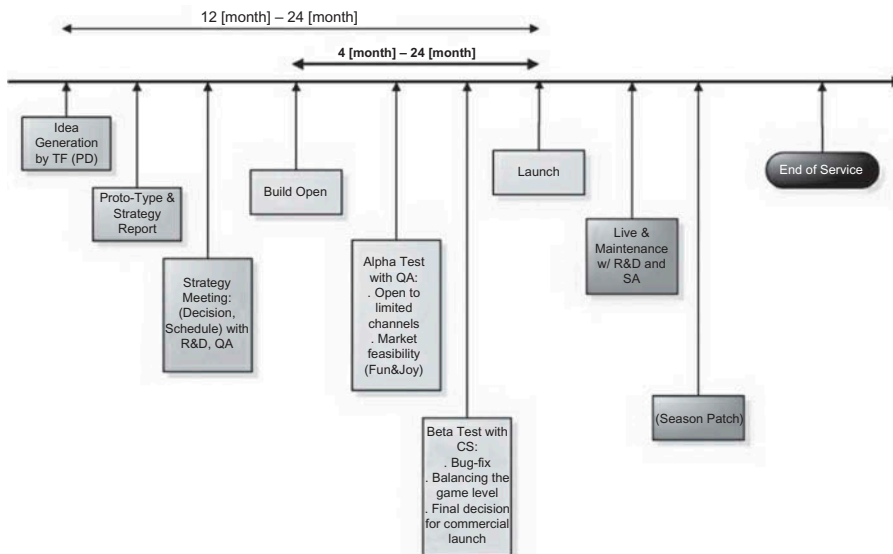


FIGURE 2 Development process of Action Puzzle Family (APF).

Tiny Farm Case

In Tiny Farm, users can have tiny farms in their smartphones and enjoy seasonal experience in the farm (Com2uS, 2014b). A user becomes a farmer and is tasked to provide care for his or her own special animals. Tiny Farm is “jam-packed with sweet surprises and endless possibilities,” supporting more than five languages. It is known as a major social network Freemium casual game, even though it started with a small group. Apart from being relatively new, the game presents a challenge to Com2uS because Tiny Farm is the first role-playing game connected with a service platform, Com2uS Hub. Unique to Tiny Farm are the comfort and ease of play, a sense of cooperation, completion with friends, and unexpected surprises during gameplay.

GMATE (the internal project name of TinyFarm) had started in 2010 with one producer (Mr. Lim), one project manager (Mr. S. Lee), two graphic designers (Mr. W. Yang and Ms. Y. Kim), and two software developers (Mr. J. Han and Mr. S. Ok). According to the e-mail interview with Mr. Lim (GMATE producer), it took around 1 year to complete the development and 1.5 years to maintain after launching the game. Most of the stages in the development process are shown below. In general, the process (except for the timelines) is the same as that of the APF game (Figure 2):

- *Development start*: The game was to be a new type of the SNG with the support of Com2uS Hub.
- *Prototyping*: The task force team made a prototype in less than 1 month that demonstrated the basic concept of the game. Most planning of the system, contents, and client/server architecture were completed during this phase. In the case of Tiny Farm, reliable communication with the Com2uS social network server was an additional challenge as articulated by Mr. Lim in his own words right before the Conclusions section.
- *Alpha test (demo)*: Tiny Farm was developed in detail, and bugs were fixed to improve user experience. Major operators in Korea, such as SKT, KTF, and LGT, participated in this alpha test. Tutorials, content development, and addition of new buildings were completed. The balancing of game level with difficulty was also partially completed during alpha test.
- *Beta test (demo)*: Beta testing was primarily meant for fixing bugs, and a customer service team of Com2uS was actively involved at this phase. Balancing the game levels and difficulty were also finalized based on the lessons learned during the alpha test.
- *Commercial launch*: After completing beta testing, GMATE (Tiny Farms) was launched on Apple AppStore on September 15, 2011, with the Com2uS Hub.

As it is mentioned in the previous section, very little documentation about the software development process was available. So, most of the information was obtained through the e-mail interview with the GMATE producer. According to Mr. Lim, “the whole development process is not one time single process, but it is the multiple cycled processes (development (improvement) → testing → evaluate loop)” (personal communication, December 2012).

Both the APF and Tiny Farm development cases suggest that the processes of the Agile model were more suitable than those of the waterfall model, even though Com2uS did not precisely determine which software development processes were actually suitable for the development of the two games. In the case of TinyFarm, the game required higher standard in the terms of software quality (the client side) because TinyFarm was the first mobile game that intercommunicated with Com2uS Hub on the top of the real-time data communication with the game server.

Com2us Hub is the game service platform that supports the unified service for mobile games and also supports activities that include new game advertisement, managing users, social networking, handling forums, and aligning various devices from one user to another (Kim, 2014). It required more complicated and reliable data communication between the client (i.e., TinyFarm on a smart-phone) and the servers (i.e., Com2uS Hub server and TinyFarm game server). According to Mr. Lim:

Every judgment of each user is fully handled by the game server, handling the overload of server during the peak access time is really vital issue. In addition, adapting Com2uS Hub is additional critical issue to be solved effectively at the same time. Since, both connections are required, reliable and real-time data communications between the client and both servers (the game server and Com2uS Hub), the software on the client must be reliable enough for testing the servers. (personal communication, December 2012)

CONCLUSIONS

From the previous description, it appears that the Agile development rather than the classical waterfall model was utilized for game software development in these two cases. Each student should put himself or herself in the role of project manager of a mobile game development company and discuss the following issues:

- What are the potential risks when a particular development process is deployed?
- What steps should be taken to mitigate these risks?
- How can relevant features of different software development processes be adapted to develop a mobile game?
- What are some additional challenges for developing mobile social network games like Tiny Farm?
- Lack of documentation prevailed in the two cases—in this context, key questions are: What are the implications of these for future software game development? Should companies utilize software that facilitates documentation without much effort?
- How does one rate the success of two games in the current market?

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REFERENCES

- Benington, H. D. (1983). Production of large computer programs. *IEEE Annals of the History of Computing (IEEE Educational Activities Department)*, 5(4), 350–361.
- Chung, J., & Park, R (2012). Mobile game industry. *Woori Investment and Security*. Retrieved from <http://www.wooriwm.com/>
- Cohen, S., Haan, U. D., & Dori, D. (2010). A software system development life cycle model for improved stakeholders' communication and collaboration. *International Journal of Computers, Communications & Control*, 5(1), 20–41.
- Com2uS. (2012). *Com2uS 2012 investor relations: Financial results and earnings*. Retrieved from <http://www.com2us.com>
- Com2uS. (2014a). *Action Puzzle Family*, Google Play Store. Retrieved from <http://bit.ly/1pFP5v9>
- Com2uS. (2014b). *Tiny Farm*, Apple iTunes. Retrieved from <http://bit.ly/1AdzLAB>
- Conboy, M., & Morgan, L. (2011). Beyond the customer: Opening the Agile systems development process. *Information and Software Technology*, 53(5), 535–542.
- Hayes, T. (2008). *Jump point: How network culture is revolutionizing business*. New York, NY: McGraw-Hill.
- Heeager, L. T. (2012). Introducing Agile practices in a documentation-driven software development practice: A case study. *Journal of Information Technology Case and Application Research*, 14(1), 3–24.
- Iansiti, M., & MacCormack, A. (1997, September). Developing products on internet time. *Harvard Business Review*, pp. 108–117.
- Iansiti, M., & MacCormack, A. (1999). *Living on internet time: Product development at Netscape, Yahoo, NetDynamics, and Microsoft*. Harvard Business School Case 9-697-052.
- Kim, C. (2012). Com2Us, company report. *KDB Daewoo Securities*. Retrieved from <http://www.dwsec-id.com/>
- Kim, S.-K. (2014). Com2uS: Mobile game pioneer. *International Journal of Teaching and Case Studies*, 5(2), 189–196.
- McAvoy, J., & Butlera T. (2009). A failure to learn by software developers: Inhibiting the adoption of an Agile software development methodology. *Journal of Information Technology Case and Application Research*, 11(1), 23–46.
- Lee, D. (2012). Com2uS creating fun games for all platforms. *Presentation of casual connect Asia 2012*. Retrieved from <http://casualconnect.org/>
- Song, E. (2012). SNG (social network games). *NetTerm*. Retrieved from <http://www.kisa.or.kr/> 41–44.
- Wikipedia. (2014a). *Agile software development*. Retrieved from <http://en.wikipedia.org/wiki/>
- Wikipedia. (2014b). *Spiral model*. Retrieved from http://en.wikipedia.org/wiki/Spiral_model
- Wikipedia. (2014c). *StarCraft*. Retrieved from <http://en.wikipedia.org/wiki/starcraft>
- Wikipedia. (2014d). *V-Model*. Retrieved from <http://en.wikipedia.org/wiki/V-Model>
- Wikipedia. (2014e). *Wireless internet platform for interoperability*. Retrieved from <http://en.wikipedia.org/wiki/WIPI>
- Wikipedia. (2014f). *Waterfall model*. Retrieved from http://en.wikipedia.org/wiki/Waterfall_model
- Zynga. (2014). *Farmville official site*. Retrieved from <http://zynga.com/game/farmville>

APPENDIX A: QUESTIONNAIRE FOR GAME DEVELOPMENT

The original questionnaire was written in Korean and translated into English for international publication purposes. The questionnaire was applied for both Action Puzzle Family (APF) and TinyFarm (GMATE).

General Questions

1. Explain the game.
2. Who are the target customers?
3. Who are the competitors?
4. How does the value proposition compare with other similar games?

Game Development Questions

5. How can the game development duration be estimated?
6. What is the process of the game development?
7. Are there any management applications for the game development?
8. General information of the game development:
 - human resources and role and responsibility,
 - development planning (schedule),
 - new features for this game, and
 - technical issues and how to resolve the issues.
9. What is the burden (bottleneck) for this game development?
 - Why?
 - How can the burden be solved?

APPENDIX B: RESEARCH NOTE

The software development process is a generally referred to as the systems development lifecycle (SDLC). Several models exist for SDLC to execute a multitude of tasks from the beginning to end of a software development project. The software lifecycle typically includes the following stages: requirements, analysis, design, construction (or coding), testing (validation), installation, operation, and maintenance (Cohen, Haan, & Dori, 2010). Each model has its pros and cons, and it is up to the development team to adopt the most appropriate model for the project. Sometimes a combination of the models may be more suitable. Two primary development models are: Waterfall and Agile.

Waterfall Development Model

The waterfall model is a classical software development process which is seen as flowing like a waterfall. The developers are to follow the phases in order (Wikipedia, 2014f):

- Requirements specification (requirements analysis)
- Software design
- Implementation and Integration
- Testing (or validation)
- Deployment (or installation)
- Maintenance

In a strictly implemented waterfall model, after each phase is finished, developers proceed to the next one. Reviews may occur before moving to the next phase, which allows for the possibility of changes (that may involve a formal change control process). The waterfall method discourages revisiting and revising from any prior phase once it has been completed (see Figure A1).

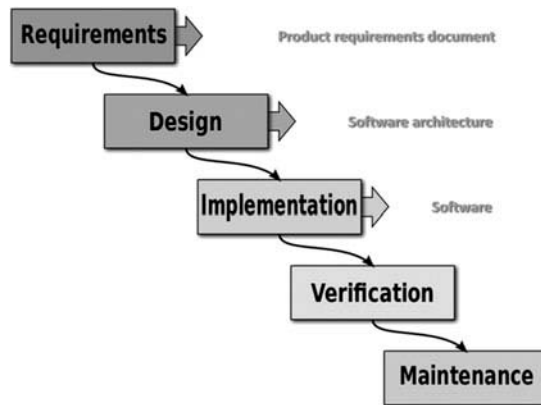


FIGURE A1 Waterfall development model (see Wikipedia, 2014f).

The waterfall development model originated in the manufacturing and construction industries; highly structured physical environments in which after-the-fact changes are prohibitively costly, if not impossible. Since no formal software development methodologies existed at the time, this hardware-oriented model was adapted for software development (Benington, 1983).

Agile Development Model

Agile software development, which is based on iterative and incremental development, is a modern software development method especially practical for startup companies (Wikipedia, 2014a). The Agile development process was introduced in the 1990s to minimize process bureaucracy by removing unnecessary milestones from administrative workloads (Conboy & Morgan, 2011). Agile processes use feedback, rather than planning, as their primary control mechanism (Heeager, 2012). The feedback is driven by regular tests and releases of the evolving software. Agile methods break tasks into small increments with minimal planning and do not directly involve long-term planning. Each iteration involves a cross functional team working on all stages of SDLC: planning, requirements analysis, design, coding, unit testing, and acceptance testing (see Figure A2).

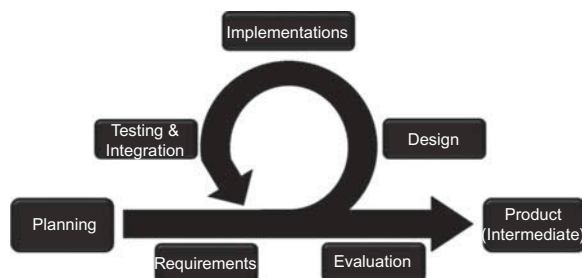


FIGURE A2 Agile development model (see Wikipedia, 2014a).

General Process of Modern Mobile Game Development

The development and rise in popularity of mobile games have many similarities to the production and distribution of films. In the film industry, it is hard to predict which movie will become a hit until the movie is actually released. In addition, movie customers seldom even know the name of the actual production company. Customers remember the names of distributors of the most popular films, not necessarily the names of producers (some companies do both). Audiences may be more interested in the cast and the director of the movie, rather than the production or distribution companies. In the case of mobile games, customers rarely care about the directors or developers. Once a mobile game hits the market, the company that develops it becomes famous even if it has produced only one hit game. DragonFlight by Next Floor is one example from the industry. This degree of customer awareness may be good news for up-and-coming developers and a new successful mobile game obliterates the previous successful game. In this fast-paced age of information technology, the fame and popularity for fresh IT products and services is short-lived. This pattern, therefore, reflects the need for fast production life cycles for movies and mobile games. Because of the similarity of market characteristics between movies and mobile games, the mobile game development should be more flexible and should have faster development cycle to survive in the current dynamic market. This need translates to employing flexible approaches that allow companies to respond to changes in markets and technologies during the development cycle (Iansiti & MacCormack, 1997).

In summary, there are four characteristics of mobile game development: First, a mobile game software industry is similar to the movie industry. Second, no one knows which game will succeed in the market. Third, it is the pure business-to-consumer interaction that characterizes the market. Fourth, short time-to-market for a mobile game is critical since market and technology are constantly changing. Stages of Com2uS mobile game development can be characterized as follows. These stages could be applied by other mobile game development companies also. The entire process has to be dynamic with the agility to reschedule subsequent stages at the end of any stage and also with ability to assign new members in the software development team:

- *Idea generation:* Ideas for new games are usually proposed by the producers in the task force team.
- *Prototyping and strategy report:* Once an idea has been recognized, the task force team makes a prototype of the game that can demonstrate the basic concept of the game.
- *Strategy meetings:* Once a demo report and the strategy report are ready, strategy meeting is set up with other task force teams (mostly development teams) and the quality assurance teams. In this meeting, the decision whether to move forward with development is made. If the answer is YES, then actual development is started. It takes several weeks to months until the build open¹ when the game starts running. A limited number of users (in the case of Com2uS, mostly game developers) can actually play the game.
- *Build open:* Once the build is opened, actual trials are initiated within the company. At this point, further development and bug-fixing are done simultaneously until the game is stable

¹The term *build open* means that the code is open for all related developers who are involved in the mobile game project. Multiple developers could access the codes by building the new features (for the commercial version) and any bug fixing.

enough to be opened up to limited channels. It may take anywhere from 2 months to 2 years before alpha testing (or a demo) takes place.

- *Alpha test (or demo)*: At this phase, the game is opened to limited channels such as mobile operators or publishers. Internally, the quality assurance team is actively involved to ensure stability and quality of the game. Additionally, this stage is the time for obtaining feedback from people outside the company. In the case of Com2uS, the company already partners with major operators in South Korea (i.e., SK Telecom, KT, and LG Telecom).
- *Market feasibility*: The fun and enjoyment factors are also determined during this phase.
- *Beta test (or demo)*: Beta testing is primarily meant for fixing bugs, and a customer service team is actively involved in the execution of this phase. Balancing the game levels and difficulty are also finalized based on the market feasibility study (user survey and corrected data) during the alpha test. The game may be terminated in this phase if it is not good enough for the current market.
- *Commercial launch*: Once all tests (alpha and beta) were completed, the game is opened to the public.
- *Live maintenance*: After launching the game, maintenance of the game (i.e., bug fixes and upgrades) is still required. In the case of SNGs, maintenance of both the game application (game software on the smartphones) and game servers (which correct the data and control the games) are to be covered.
- *Season patch (optional)*: During special seasons such as Christmas or New Year's Day, seasonal patches may be launched by AppStore and Google Play. Special skins (e.g., look and feel, colors, images) and promotions are created for seasonal patches.

In the software development process, the classical process models such as waterfall model, V-model² and spiral model³ may not be applicable for mobile game development. Mobile game development requires flexibility toward external environments, which in itself is a feature of the gaming industry. The whole development process may range from one to two years. Com2uS mobile game process has the various phases (Figure 2).

Conclusions

The software development process is often considered as a subset of systems development life-cycle for developing software products and Agile process (Wikipedia, 2014a) introduces the idea of simplicity. Agile software development never generates more than what is necessary and never produce documents attempting to predict the future. In the case of mobile game software development, Agile software development fits well in terms of a flexible approach to engage customer demands astutely. That reason is why the Agile software development process adapts well to modern mobile games, in particular for social network- based casual freemium games such as TinyFarm. As part of a series of case teaching materials, this research helps students understand the modern software development process and the characteristics of most advanced technology

²The V-model is considered an extension of the waterfall model in which the process steps are bent upward after the coding phase, to form the typical V shape (see Wikipedia, 2014d).

³The spiral model is another extension of the waterfall model which can be described as a process model generator based on the risks of a project (see Wikipedia, 2014b).

industries. It also provides insights on modern software development for technical managers and/or project managers who are involved in the field. On another note, although it shows that the processes of the Agile development method would be more suitable than the waterfall method for both games, Com2uS might also consider the waterfall method for the development process of its game project, as TinyFarm requires more reliable and real-time communication with its server. The Agile development process is very practical from the position of modern software development, but it is not a panacea for every modern rapid develop-cycle software development (McAvoy & Butlera, 2009). As case teaching material, this case research enables students to understand the modern software development process and the characteristics of most advanced technology industries. It also provides insights on modern software development for technical managers and/or project managers who are involved in the field.

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